MIL-E-1/199A
31 January 1968
SUPERSEDING
MIL-E-1/199
20 May 1953

MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 6SC7, 6SC7GTY, AND 12SC7 1/
a b c

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin triode, high Mu

Outline	---	a, c	8-1 (EIA)
		b	9-7 or 9-39 (EIA)
Base	---	a, c	B8-21
		b	B8-6 or B8-46 (low-loss phenolic)
Envelope	---	a, c	MT-8
		b	T9
Cathode	---	a, b, c	Coated unipotential

Base connections:

Pin No.	---		1	2	3	4	5	6	7	8
Element	---	a, c	sh	2p	2g	1g	1p	k	h	h
Element	---	b	nc	2p	2g	1g	1p	k	h	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:		Ef	Eb	Ec	Ehk	Alt
Unit:		V	Vdc	Vdc	v	ft
Maximum:	a, b	6.93	275	---	100	10,000
Minimum:	a, b	5.67	---	---	---	---
Maximum:	c	13.86	275	---	100	10,000
Minimum:	c	11.34	---	---	---	---

TEST CONDITIONS:

a, b,	6.3	250	-2	---	---
c	12.6	250	-2	---	---

REQUIREMENT OR TEST:

GENERAL

3.2	Qualification - Required	E-50.3	Preheating Ⓐ
E-50.2	Holding period	5.	Preparation for delivery

1/ To identify immediately those tests that are applicable to a given type or to several types; tube types are designated by letters.

Ⓐ denotes technical changes only.

6SC7, 6SC7GTY, 12SC7
a b c

FSC 5960

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	a, b, c	See notes 1 and 3	0.65	II	Ic	0	-1.0	μAdc
1256	Electrode current (anode)	a, b, c	See note 2	0.65	II	Ib	1.2	2.8	mAdc
1321	AC amplification	a, b, c	Esig = 0.2 Vdc; Ebb = 100 Vdc; Ecc = 0 (see note 2)	0.65	II	Ep	7.5	---	Vac
Ⓐ1246	Audio frequency noise	a, b, c	Esig = 70 mVac; Rp = 0.2 Meg (see notes 1 and 4)	0.65	II	---	---	---	---
1231	Emission	a, b, c	Eb = Ec = 30 Vdc (see notes 2 and 3)	0.65	II	Is	30	---	mAdc
1201	Short and discontinuity detection	a, b, c		0.4	II	---	---	---	---
Appendix D, 20(a), 30	Visual and mechanical inspection criteria	a, b, c		---	---	---	---	---	---
Appendix D, 20(b)	Dimensions	a, c	Outline 8-1; dimensions A, B, C, D, L, and M	1.0	I	---	---	---	---
		b	Outline 9-7 or 9-39; dimensions A, B, C, and D	1.0	I	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes	a, b, c		4.0	L6	R	10	---	Meg
1031	Low-frequency vibration	a, b, c	Rp = 2,000 ohms (see note 1)	6.5	L6	Ep	---	150	mVac
1111	Base pin solder depth	a, b, c		6.5	L6	---	---	---	---
1301	Heater current	a, b		6.5	L6	If	275	325	mAdc
		c		6.5	L6	If	138	162	mAdc
Ⓐ1336	Heater-cathode leakage	a, b, c		6.5	L6	Ihk	---	20	μAdc
1101	Secureness of base and base insert	a, b, c		6.5	L6	---	---	---	---
1306	Transconductance	a, b, c	See note 2	6.5	L6	Sm	1,000	1,650	μmhos
1316	Amplification factor	a, b, c		6.5	L6	Mu	55	85	---
1331	Direct-interelectrode capacitance	a, c	See note 2	6.5	L6	Cgp	---	2.4	pf
						Cin	1.2	2.8	pf
						Cout	2.0	4.0	pf
1105	Permanence of marking	a, b, c		---	---	---	---	---	---

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
4.7 and 1501	**Life-test provisions (intermittent)**	a, b, c	Group A; Ehk = 100 V (see note 1)	---	---	---	---	---	---
4.7.3	**Life-test end points (intermittent) (500 hours)**	a, b, c	AC amplification or Transconductance	--- ---	--- ---	Ep Sm	6.5 860	--- ---	Vac μmhos

NOTES:

1. Test units connected in parallel.
2. Test each unit separately.
3. This test to be performed at the conclusion of the holding period.
4. The rejection level shall be set at the VU meter reading obtained during calibration. Ⓐ
5. The tubes described by this TSS are interchangeable with the tubes procured to the previous issue. Ⓐ

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Army - EL

(Project 5960-2199-14)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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